CSE 491 Project Proposal

1. Title and summary

Title: A Human-in-the-Loop Knowledge Graph Generation Engine for Preventing Hallucination and Schema Drift in LLMs: An Application on Hadith Isnad Chains

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Summary. Large language models can turn free text into knowledge graph facts, but they add facts the source does not contain and they drift away from the schema they were given. GiraffeGraph is an engine in which the model only proposes. Each proposed fact carries a quotation from the source, is checked by deterministic rules, and reaches the graph only through a recorded human decision or a recorded acceptance policy. The design rests on one claim: what such a system has to store is the decision of a person, and what can be derived from the source is derived again. Our case study is the hadith isnad: the chain of narrators that precedes a hadith text. By the end of CSE 491 we will deliver a working prototype that processes a small set of Arabic hadiths end to end and reports, against hand-labelled data, how many errors each check catches. Two supporting studies test the claim itself: a simulation of reusing narrator-identity decisions on labelled chains, and an experiment in a second domain. Entity resolution of narrators inside the engine, the full experiment and the question-answering layer are planned for Senior Design Project II.

2. Problem and motivation

When an LLM extracts a narrator chain from a hadith, four things go wrong:

1. **Unsupported facts.** The model outputs a narrator or a link that the text does not contain, or it silently rewrites a name it did find.
2. **Schema drift.** The same relation appears as `NARRATED_FROM` in one run and under another name in the next, and property names vary between runs.
3. **Identity confusion.** One narrator is written in several ways, and several narrators share one short name. Merging two different people changes how the chain is judged.
4. **No trace.** After the graph is built, nobody can tell which sentence a fact came from or who accepted it.

These matter in this domain because the reliability of a hadith is judged from its chain. A single wrong narrator changes the verdict, so a graph that is mostly right is not usable by a scholar.

Target users. Researchers and students in hadith studies who need to browse and query narrator chains, and engineers who build knowledge graphs from text in other domains where each fact has to be traceable to its source.

What we already observed. A pilot run on 30 September 2026 on 50 hadiths (327 reference narrator mentions, one model, scored against the rule-generated labels of the Sanadset corpus, loose matching) gave three results that shape this proposal:

- A rule-based parser reached F1 0.879 and an LLM with a schema-only prompt reached 0.865. An LLM with an isnad-aware prompt reached 0.935. Raw extraction accuracy is therefore not where the project can contribute most.
- Checking that the model's quotation occurs in the source caught nothing (605 of 605 quotations were verbatim). Checking that the extracted name occurs inside the model's own quotation

flagged 32 of 318 candidates.
- The model's self-reported confidence separated right from wrong candidates poorly (AUC 0.626).

These numbers are comparisons between methods on rule-generated labels. They are not accuracy figures, and one of the aims of the project is to replace them with measurements on hand-labelled data.

The claim behind the design. A candidate that a parser or a model extracts from a text can be extracted again, so it is treated as derived: it is checked, not trusted, and it does not have to be kept. An acceptance, a correction or the identification of a narrator by a person cannot be derived from any text and has to be stored once, with who decided and when. Two exploratory studies of 13 October 2026 looked at the second half. Both are simulations: no person took part.

- On the 27,056 test chains of AR-Sanad [4] (150,904 narrator mentions), a stored identity decision that is reused whenever the same written name appears is wrong for one mention in ten. When the decision is tied to its context only after a deterministic rule on generations and death years has seen a contradiction, the silent error falls to 0.014 per mention with about half the reviewer questions that a fixed context key needs. The reviewer is simulated from the labels of the data set.
- In a second domain, a coding agent that answers questions about the engine's own source code answered questions about why the code is the way it is 0% to 4% correctly from the code alone and 88% to 100% correctly with a ledger of decisions (12 questions, two runs each). It answered structural questions 94% to 98% correctly with no documentation when it had enough tool calls.

3. Objectives, scope and success criteria

Objective. Build an engine that turns hadith text into a narrator-chain graph in which every fact is tied to a source quotation and to a recorded decision, and measure how much each validation step reduces extraction errors. Test the claim behind the design, that decisions have to be stored and derivable facts do not, on the engine's own task and in one other domain.

In scope for CSE 491 (prototype):

- Loading Arabic hadiths, one hadith per source unit, from Sanadset 650K [1].
- Rule-based separation of isnad and text, and splitting of the isnad into narrator mentions.
- LLM extraction of narrators and `NARRATED_FROM` links, one call per chain, constrained to a fixed schema.
- Three deterministic checks: the type exists in the schema; the quotation occurs in the source; the extracted value occurs in the quotation. One domain rule that needs no biographical data: the Prophet is not a narrator.
- A review screen where a person accepts, rejects or edits a candidate, with who decided, when, and how long it took.
- A canonical graph in which the same two narrators can be linked in different hadiths without one link overwriting the other.
- A scoring script and a hand-labelled set of 50 hadiths.
- Two supporting studies, reported with their scripts and limits: the simulation of a ledger of narrator-identity decisions on labelled chains, and the coding-agent experiment. Neither changes the engine.

Out of scope for CSE 491 (planned for Project II unless stated otherwise):

- Linking narrator mentions to an authority list of known narrators (entity resolution), and a ledger of identity decisions inside the engine. CSE 491 has the simulation only.
- Domain rules that need biographical dates (contemporaneity, meeting, generation).
- The comparison of graph-based and text-only retrieval for question answering.

- Multi-model experiments and a labelled set larger than the 50-hadith calibration set.
- Turkish hadith text. We found no copyright-free Turkish source with full chains.
- Judging the authenticity of hadiths. The system records chains; it does not grade them.

Success criteria. The prototype is tested against these in week 13.

#	Criterion	How it is measured
S1	A set of 100 hadiths passes from upload to canonical graph without manual intervention outside the review screen	End-to-end run, logged
S2	100% of facts in the canonical graph have a source quotation and a decision record (human or policy)	Automated test over the graph
S3	Precision, recall and F1 are reported for four conditions (rule-based parser; LLM; LLM with checks; LLM with checks and human review) on the 50 hand-labelled hadiths, under strict and loose matching, with hadith-level bootstrap confidence intervals. For each check, the table also gives how many mismatched proposals it flags on its own, next to the number the model's confidence flags when the same share of proposals is sent to review	Scoring script, results table
S4	Inter-annotator agreement on the 50 hadiths and mean labelling time per hadith are reported	Two independent annotators, mutual F1
S5	For the claim "decisions are stored, derivable facts are derived again", the report states what was measured for it and against it, and every number of the two supporting studies is reproduced by a script in the repository	Re-run of the scripts from a clean clone

4. Similar systems and background

Work	What it provides	How quality was measured	What is missing for our purpose
Sanadset 650K [1]	650,986 hadith records from 926 books with narrator tags	Not reported; tags were produced with regular expressions	No human check of the labels
SemanticHadith [2], [12]	Ontology and knowledge graph for the six canonical collections (34,458 hadiths); entities extracted with a custom NER model; narrator chain segments are modelled as classes	NER micro F1 97%; three experts checked a random sample of 100 instances per collection	No LLM extraction and no decision per fact; ambiguous names are excluded or left for manual review, and an expert validation framework is listed as future work

Work	What it provides	How quality was measured	What is missing for our purpose
Çinici [13]	LLM with retrieval turns one volume of a classical biographical work into a structured dataset of 271 scholars	The author compared the first 100 records with the source; no figure reported	Biographies, not chains; one evaluator; no graph or review workflow
Muther and Smith [3]	Expert-annotated isnad boundaries in 50 classical texts	Strict F1 0.254, partial F1 0.835; experts agree exactly on the end point in 53% of cases	Finds isnad spans, not individual narrators
AR-Sanad 280K [4]	27,056 real chains with narrator identities, 18,298 narrators	Micro F1 83.5, but 60.6% of chains contain at least one wrong narrator	Starts from already separated names, not from text
Oumer [5]	Fine-tuned models for narrator and relation extraction on 223,478 hadiths	Narrator F1 0.98, relation F1 0.95 (figures from the author's model repository)	Needs large labelled training data; no provenance or human review
Rezwan [6]	LLM-assisted corpus of 1.2M narrations	Six experts, 0–10 scores; no agreement statistic	Scores are not comparable across studies
Wirahman [7]	Links 185,216 narrator name variants to biographical databases	Not measured; the author states no labelled reference exists	Fully automatic, no human decision
Text2KGBench [8]	Benchmark for ontology-driven KG extraction with LLMs	Precision 0.38 with subject hallucination 0.17 on one dataset	General domain; shows that "appears in the text" does not imply "correct"
Tsaneva et al. [14]	Nine workflows that combine LLM validators and people in the validation of a scientific KG	F1 against three experts; number of triples people judged	People see all triples, low-support ones or validator disagreements; modules are not evaluated in isolation
KARMA [15]	Multi-agent LLM enrichment of a biomedical KG with validation agents	Ablation of agents, scored by an LLM judge	No external reference; agents, not single checks
ODKE+ [16]	Production extraction with a second LLM as grounding judge	Grounding and corroboration effects reported without a method	No deterministic check of the extracted value

A recent critical review of the field [9] lists weak benchmark comparability and sparse expert validation as open problems. General frameworks for LLM-based graph construction exist as well [10], [11].

Position of this project. A hadith knowledge graph, its ontology and narrator linking already exist, and fine-tuned models already extract chains accurately. We build on that work for the schema. Outside the hadith domain, validation between an extractor and a graph has been compared as whole workflows, ablated agent by agent with an LLM judge, and routed to people by confidence or by disagreement [14]–[16]; we do not claim that idea. We did not find a check that the extracted value occurs in the model's own quotation, reported as its own error class, or a measurement of what each deterministic check catches on the same candidates against an external reference, compared with the model's own confidence at the same review share. We also did not find a measurement of how often a stored narrator-identity decision can be reused and at what error. We do not claim that the mechanism we simulate, tying a correction to the context in which an earlier answer failed, is new. Our search was not systematic, so this is a statement about what we found.

5. Preliminary requirements

Functional

ID	The system shall
F1	let a user define node types, relation types and their properties, and reject identifiers that the graph database cannot store
F2	ingest a hadith as one source unit with a stable identifier and content hash
F3	split a hadith into isnad and text and return narrator mentions with character offsets
F4	request candidates from an LLM for one chain at a time, constrained to the schema
F5	reject a candidate whose type is not in the schema without discarding the other candidates of the same call, and record the unknown type
F6	mark a candidate for review when its quotation does not occur in the source
F7	mark a candidate for review when its extracted value does not occur in its quotation
F8	present candidates with their quotation and check results, and let a reviewer accept, reject or edit them
F9	record for every decision the reviewer, the time and the duration
F10	write accepted facts to the canonical graph so that links between the same two narrators in different hadiths stay distinct
F11	display the chain of a hadith and the source quotation of each link
F12	score system output against a reference file and produce precision, recall and F1
F13	mark a candidate for review when it names the Prophet as a narrator

Non-functional

ID	Requirement	Verification
N1	No fact reaches the canonical graph without a decision record	Automated test (S2)
N2	Re-running a commit does not create duplicate nodes or links	Integration test
N3	The backend test suite runs without a database or network access in under one minute	CI
N4	A new user can start the system from the README alone	Tested on a clean machine in week 14
N5	No API keys or passwords are stored in the repository	Repository scan in CI

6. Method and technology plan

Approach. The engine is a pipeline with a gate at each stage:

```
hadith text → isnad segmenter → LLM proposal (per chain) → checks → review → canonical
graph
                                                          schema / quotation / value
decision record
```

The model has no write access to the graph. A candidate becomes a fact only after it passes the checks and receives a decision.

Measurement. We add the stages one at a time and score each configuration against the hand-labelled set, so the contribution of each stage can be read from the table. The matching criterion is fixed before measurement: strict (normalised strings equal) and loose (equal, substring, or token F1 of at least 0.60), with maximum-weight matching between system and reference mentions. The hadith is the unit of analysis.

Labelled data. Two group members label the same 50 hadiths independently, following a written guideline, without seeing system output. The person who writes the checks does not label. The domain expert reviews the guideline before labelling starts and decides where the two annotators disagree. The expert does not review system output, so that the person who sets the reference does not also judge the system.

Existing code. GiraffeGraph Core already implements schema versioning, ingestion, LLM extraction through an OpenAI-compatible provider, candidate validation, a review queue and commit to Neo4j. A first version of the checks in F5–F7, the decision record in F9 and the link identity in F10 was written in October with AI coding assistants and is covered by automated tests. It has not been reviewed by the group and has not been used by a reviewer. Reviewing it, testing it with a person in the loop and measuring it against hand-labelled data is the main work of this term.

Technology

Layer	Choice	Reason
Backend	Python 3.12, FastAPI	Existing code base; team experience
Operational store	PostgreSQL 16	Runs, candidates, review items and decisions are relational
Canonical graph	Neo4j 5	Path queries over chains
Internal dashboard	Next.js 15	Existing review interface
LLM access	OpenAI-compatible HTTP API	Lets us switch between hosted and open models
Development	uv, pytest, ruff, Docker Compose, GitHub Actions	Reproducible setup and CI
Data	Sanadset 650K	Arabic text with full chains under an open licence (CC BY 4.0 on the original record; the mirror we download states CC0)

A decision table with the alternatives considered for each choice will be part of the midterm report.

7. Work plan

Week	Dates	Work	Output
5	12–18 Oct	Proposal	This document
6	19–25 Oct	Architecture, data model, three screen sketches;	Diagrams; guideline v1; supervisor meeting 3

Week	Dates	Work	Output
		guideline reviewed with the domain expert	
7	26 Oct – 1 Nov	Technology decision table; labelling of 50 hadiths by two annotators	Decision table; labels; time per hadith
8	2–8 Nov	Midterm report and presentation; agreement figures; disagreements decided by the expert	Report, slides, dated plan; reference set
9	9–15 Nov	Group review of the existing engine code; one hadith end to end; supervisor and course staff added to the repository	Review notes; running skeleton; supervisor meeting 4
10	16–22 Nov	Segmenter, per-chain extraction and the three checks reviewed and tested by the group; the domain rule added	Core components with tests; repository link on Teams
11	23–29 Nov	Review screen used by a group member on the labelled set; decision record and chain view tested	Clickable workflows; supervisor meeting 5
12	30 Nov – 6 Dec	Integration on 100 hadiths, scoring script, deployment	First tagged version
13	7–13 Dec	Test against S1–S4, fixes, list of known limitations	Results table
14	14–20 Dec	Final report, README, technical documentation	Report and repository
15	Final period	Prototype demo and plan for Project II	Presentation

Division of work

Member	Main responsibility
Yahya Efe Kuruçay	Architecture, validation checks, decision record, scoring script
Deniz Büyükşahin	Labelling guideline, annotation (annotator 1), review screen
Alperen Ulukaya	Isnad segmenter, annotation (annotator 2), chain view

All members commit their own work under their own accounts.

8. Expected outputs and risks

Outputs at the end of the term

- A working prototype and its repository with README, tests and CI. Engine, documentation and data are in one private repository; the engine is published under an open-source licence at the end of the project.
- A labelling guideline and 50 hadiths labelled by two annotators, with agreement figures.

- A results table for the four conditions in S3.
- The two supporting studies with their scripts, result files and stated limits.
- Final report, presentation and a plan for Project II.

Risks

Risk	Early sign	Response
Labelling takes longer than planned	Time per hadith in week 7	Reduce the set; label in a spreadsheet instead of building a tool
Annotators disagree	Low mutual F1 in the first round	Add boundary rules to the guideline and repeat the round
Development starts late	No running skeleton by mid-November	Simplify the interface; keep checks and scoring
Scope grows again	A new planning round with no working component	Return to the out-of-scope list in section 3
LLM cost or access	Calls fail or the budget is used up	Use an open model through the same API; reduce the set
Licence of additional labelled data cannot be confirmed	No written confirmation	Use Sanadset and our own labels only

Use of AI assistants

AI assistants were used for the literature search, for the pilot analysis of 30 September 2026, for code review of the existing engine, for the first version of the checks, for the two exploratory studies of 13 October 2026 and for drafting this document. Bibliographic records were retrieved from Crossref, arXiv and the ACL Anthology, not written from memory. The group is responsible for every statement here.

References

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